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Master Thesis

結合多種相似度指標與深度學習模型以預測 NBA 新秀

三年的表現

Three-Year Performance Prediction for NBA Rookies Using an
Ensemble of Similarity Measures and Deep Learning Models

研究生：孫瑋澤 (Sun, Wei-Tse)

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中華民國 一一四年七月

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誌 謝

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孫瑋澤 於

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結合多種相似度指標與深度學習模型以預測 NBA 新秀三年的表現

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摘 要

本論文提出兩套預測 NBA 新秀在生涯四到六賽季整體平均表現的方法。第一套為結合多重相似度指標，對目標球員與歷史球員分別計算加權均方誤差 (WMSE)、峰值訊雜比 (PSNR)、全域畫質指標 (UQI) 與影像資訊保真度 (VIF) 四種相似度，再以最小平方法估計最佳權重，計算成綜合分數。之後，選擇分數最高的五位歷史球員，依權重加權其生涯四到六年的平均數據，作為目標新秀的預測值。第二套為深度學習模型，透過多層感知機 (MLP) 從新秀前三年的數據學習，直接產生預測值。最後，本研究結果與線性迴歸及支持向量迴歸比較 (SVR)。顯示相似度模型在預測得分上，相對誤差 (MAPE) 為 0.0859，四種方法中第二低。而多層感知機在預測助攻上，誤差為 0.1351，是四種方法最低。這些顯示了提出的兩套方法在預測 NBA 球員表現上的可行性。

關鍵字：NBA 新秀、加權均方誤差、峰值訊雜比、全域畫質指標、影像資訊保真度、深度學習、多層感知機

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Abstract

In this thesis, we propose two methods for predicting the average performance of NBA rookies in their fourth to sixth professional seasons. The first method is an ensemble similarity measures model. For each target rookie and every historical player, four complementary similarity measures are computed: Weighted Mean Squared Error (WMSE), Peak Signal to Noise Ratio (PSNR), Universal Quality Index (UQI), and Visual Information Fidelity (VIF). The least squares method is then applied to estimate the optimal weights, combining these measures into a single ensemble score. The five historical players with the highest ensemble scores are selected, and their average performance of the fourth to sixth seasons is combined to generate the rookie's prediction. The second method is a deep learning model that uses a Multilayer Perceptron (MLP) to learn from the first three years of performance data and generate predictions for the target rookie. Finally, the results are compared with Linear Regression and Support Vector Regression (SVR). The similarity model achieves a Mean Absolute Percentage Error (MAPE) of 0.0859 in predicting points, it is the second lowest among the four methods. Meanwhile, MLP achieves the lowest error in predicting assists, with a MAPE of 0.1351. These results demonstrate the feasibility of the proposed methods in predicting NBA player performance.

Keywords: NBA rookies, Weighted Mean Squared Error, Peak Signal-to-Noise Ratio, Universal Quality Index, Visual Information Fidelity, deep Learning, Multilayer Percep-

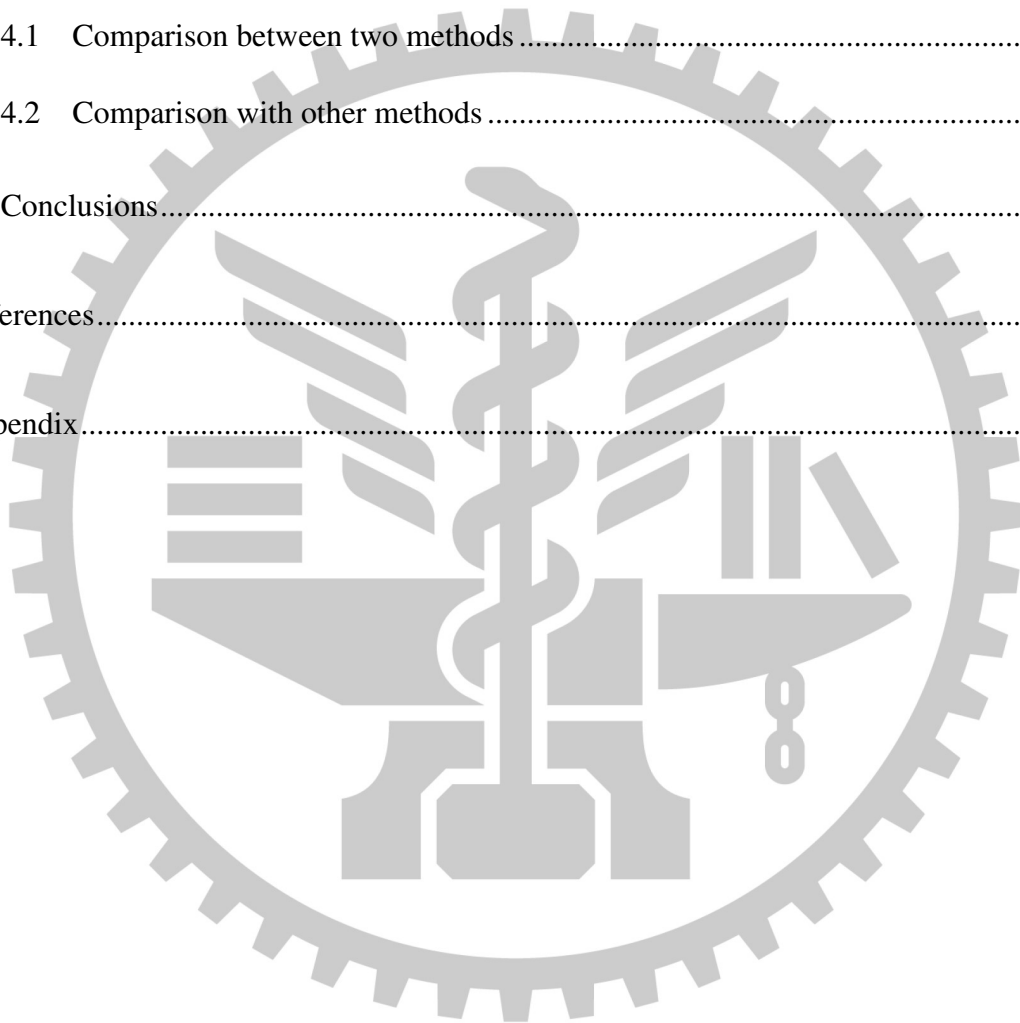
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Chapter 1. Introduction

Since its establishment in 1946, the National Basketball Association (NBA) has been recording player statistics. In recent years, data has become an indispensable tool for teams' decision-making and game outcomes. It not only quantifies a player's technical performance but also provides deeper insights into tactical efficiency, team composition, and potential evaluation. Among these, player performance prediction has emerged as a key area in sports analytics. By analyzing a player's past performance, physical attributes, and game context, we can forecast future trends in metrics such as points, rebounds, and assists. These predictions are widely used for draft evaluation, post-injury recovery planning, rotation management, and contract negotiations. Prediction techniques have also evolved from traditional regression analysis to machine learning and similarity measure models, enabling teams to make more accurate and scientific strategic decisions, thereby enhancing their competitive edge.

Approximately 60 rookie players enter the NBA each year. As unknown assets, whether they will develop into future superstars remains a widely discussed topic. Therefore, predicting rookie performance has become an important tool in player evaluation.

In this thesis, we focus on high-potential rookie players who entered the NBA between 2016 and 2018 and have made at least one NBA All-Star appearance. Based on their 16 different average states in the first three seasons of their careers (see Appendix Table A.1), we apply two methods: (1) an ensemble of similarity measures model and (2) deep learning with a Multilayer Perceptron (MLP) to predict their average performance in seasons four through six. All player data used in this thesis is sourced from Basketball-Reference [1].

For the ensemble of similarity measures model, we modify four image-based similarity methods, including Weighted Mean Squared Error (WMSE), Peak Signal to Noise Ratio (PSNR), Universal Quality Index (UQI), and Visual Information Fidelity (VIF), to compute the similarity between rookies and historical players based on their traditional stats. Then, we use a linear combination to get a final score ranging from negative infinity to 100, which represents the overall similarity. Finally, we apply this score to compute the prediction for the target rookie by the weighted average of the most similar historical players' future performance.

For the MLP model, we use the data of the first three seasons of historical players as our training data to learn the relationship with their performance in the next three seasons. Since we have limited data and may have a risk of overfitting during training, we added regularization to improve the generalization ability. The output vector is the predicted values for our target rookie eventually.

After getting predictions from both methods, we evaluate the accuracy using relative error based on the L2-norm. The ensemble similarity model and MLP method reach relative errors of 0.1243 and 0.1205, and the standard deviations are 0.0625 and 0.0712, respectively. Both methods show good prediction performance in PTS and AST.

The arrangement of this thesis is outlined as follows: Chapter 2 provides a review of related work on player performance. Chapter 3 describes the methods in this thesis, including the ensemble similarity measures model and MLP. Chapter 4 presents the experimental prediction results and discusses the advantages and weaknesses, and why these weakness can happen and how to deal with them. Conclusions are summarized in Chapter 5.

Chapter 2. Related Work

NBA data prediction has been developing for many years. In [2], each season of every player with more than three years of performance was used as an input dataset. Using Linear Regression and Support Vector Regression (SVR), they predicted the player's PTS, REB, and AST for the next season. The results showed that SVR is better than Linear Regression in predicting these three features.

In Another study [3], they wanted to use sparse and irregular data for predicting the Win Shares (WS) and the Box Plus/Minus (BPM). They used two methods for their predicted projection. One is Regularized Optimization for Prediction and Estimation with Sparse (ROPES) and the other is principal components analysis through conditional expectation (PACE). By comparing with Career-Arc Regression Model Estimator with Local Optimization (CARMELO), which for a player of interest, identifies similar players throughout NBA history and uses their careers to forecast the future player's activity [4], the results showed that PACE is more accurate than ROPES in values. However, ROPES gives a more complete prediction interval and helps get the uncertainty of future performance, making it more suitable for applications in reality.

In recent years, since the development of technology, computers can help us in completing many works. Machine learning make this thing more efficient. By repeatedly training on large amounts of data, machine learning models can complete analytical works in few seconds that would traditionally take hours or even days. It brings efficiency and accuracy to industries such as business, healthcare, and scientific research that we have never seen before. Machine learning is also applied to NBA data prediction.

In [5], they wanted to predict the daily line-up for Daily fantasy sports by using 14 machine learning. They took 201 NBA players in 2020-2021 season and used their last ten seasons' and last three seasons' performance for the dataset. The result showed that the Voting Meta-Model and the Random Forest had strong performance in predicting Fantasy Points. The daily line-up in this study ranked in the top 18.4%, and profitable line-up reached the top 23.5%. This unique approach proves the proposed methodology ' s effectiveness and emphasizes its profitability, as the optimizer process delivers positive results.



Chapter 3. Methodology

3.1 Players selection

3.1.1 Rookie players

It is not easy to determine whether a rookie player has great potential. However, popularity usually correlates with performance on the court. The better players' performance, the more popular they are. Being selected for the annual All-Star Game is a big recognition for rookie players. It means that they can play well and consistently. Therefore, we see All-Star selection as an important tool for judging a rookie's potential.

In addition, to verify the accuracy of our prediction methods, we need to observe players' performance over at least six seasons. Therefore, we chose rookie players who entered the NBA between 2016 and 2018 as our target players. Based on the above, in this thesis, the target rookie players selected must satisfy the following conditions:

- Drafted the NBA in between 2016 and 2018 and playing for at least six seasons
- Having at least one NBA All-Star appearance in his first six seasons.

There are 17 selected rookie players. (see Appendix Table A.2)

3.1.2 Historical players

In recent years, the NBA has big change in playing style. In [6], it said that after the Golden State Warriors won the championship in 2014 by using many three-point shooting tactics, many other teams began to follow this way. Since 2015, the overall number of three-point attempts in the NBA increased significantly. Three-point shooting has gradually become an important weapon for winning. In the 2024 season, the Boston Celtics averaged 48.2 on three-point attempts per game, it was the highest in the league. Celtics won the championship in that year, it shows that three-point shooting plays a key role in modern basketball.

Therefore, in order to match the modern style of play, we selected historical players starting from the 2015 season. On the other hand, to meet our goal of predicting high-potential rookies, we also put the condition of having at least one All-Star appearance. Based on the above, in this thesis, the historical players selected must satisfy the following conditions:

- From the 2015-2016 to 2023-2024 NBA seasons, having at least one NBA All-Star appearance.

There are 53 selected historical players. (see Appendix Table A.3)

3.2 Method 1: Ensemble similarity measures model

Inspired by [4], the CARMELO projection gives us the idea of using similarity to predict player performance. However, unlike the original study, our goal is to predict a player's performance in traditional statistics rather than predicting a weighted computed value.

For each target rookie player, we let their data be a 16-dimensional vector X , and compare it with historical players Y using the following four similarity measures.

3.2.1 Normalization

Before computation, we applied normalization as a preliminary step to unify the value of different features. We want to let every feature has the same range $[0, 1]$. For both rookie players and historical players, denoted as X_i and Y_j , the k -th feature after normalization is given by:

$$x'_{i,k} = \frac{x_{i,k}}{M_k}, \quad i = 1, 2, \dots, 17, \quad k = 1, 2, \dots, 16$$

$$y'_{j,k} = \frac{y_{j,k}}{M_k}, \quad j = 1, 2, \dots, 53, \quad k = 1, 2, \dots, 16$$

where:

- M_k is the maximum value of k -th feature among all rookie players and historical players.
- $x_{i,k} \in X_i, y_{j,k} \in Y_j$.

3.2.2 Similarity measures

When computing the similarity of two vectors, we use Weighted Mean Squared Error (WMSE) and Peak Signal to Noise Ratio (PSNR) to consider the distance difference between two vectors. Additionally, we want to consider the structural difference between two vectors, we use Universal Quality Index (UQI) and Visual Information Fidelity (VIF). By combining both distance and structural aspects, this method has a more complete consideration in the similarity between vectors.

Therefore, suppose that we have two vectors X'_i and Y'_j , the similarity is computed as follows.

Weighted mean square error (WMSE)

Mean Square Error is a common method for measuring similarity and it can be used in many situations. In this thesis, we apply different weights to each feature to get the Weighted Mean Squared Error (WMSE). For WMSE, the formula is defined as follows:

$$WMSE(X'_i, Y'_j) = \frac{\sum_{k=1}^{16} W_k (x'_{i,k} - y'_{j,k})^2}{\sum_{k=1}^{16} W_k}$$

where:

- W_k is the weight of different features. (see Appendix Table A.1)
- $x'_{i,k} \in X'_i$ and $y'_{j,k} \in Y'_j$.

Peak signal to noise ratio (PSNR)

PSNR is used to verify the differences between images. In this thesis, we apply PSNR to compute the similarity between vectors. For PSNR, the formula is defined as follows:

$$PSNR(X'_i, Y'_j) = 20 \log_{10} \left(\frac{MAX}{WMSE(X'_i, Y'_j)} \right)$$

where $WMSE(X'_i, Y'_j)$ is Weighted Mean Square Error as mention before, and MAX is the maximum possible value of the vector elements. Since all vectors in this study are normalized with their values constrained within the range $[0, 1]$, we set $MAX = 1$.

Under this setting, PSNR reflects the distance difference between two vectors, where a higher PSNR value means two vectors are more similar, and if two vectors are the same, the value is 33,7105 here. Compared to WMSE, PSNR uses a logarithmic function, making it more sensitive to small changes in error. This gives a more intuitive and stable way to measure similarity.

Universal quality index (UQI)

UQI [7], it is used to evaluate the structural similarity between two images. Considering overall similarity from three aspects, luminance, contrast, and structure. In this thesis, we apply UQI to measure the similarity between vectors.

1. Luminance:

$$\frac{2\bar{X}'_i\bar{Y}'_j}{(\bar{X}'_i + \bar{Y}'_j)^2}$$

2. Contrast:

$$\frac{2\sigma_{X'_i}\sigma_{Y'_j}}{(\sigma_{X'_i}^2 + \sigma_{Y'_j}^2)}$$

3. Structure:

$$\frac{\sigma_{X'_iY'_j}}{\sigma_{X'_i}\sigma_{Y'_j}}$$

For UQI, the formula is defined as follows:

$$UQI = (Luminance) \cdot (Contrast) \cdot (Structure)$$

$$UQI(X'_i, Y'_j) = \frac{4\sigma_{X'_iY'_j}\bar{X}'_i\bar{Y}'_j}{(\sigma_{X'_i}^2 + \sigma_{Y'_j}^2)(\bar{X}'_i^2 + \bar{Y}'_j^2)}$$

where:

- $\bar{X}'_i$ and $\bar{Y}'_j$ are the mean of two vectors X'_i and Y'_j .
- $\sigma_{X'_i}$ and $\sigma_{Y'_j}$ are standard deviations of two vectors X'_i and Y'_j .
- $\sigma_{X'_iY'_j}$ is covariance of X'_i and Y'_j .

The range of UQI is $[-1, 1]$, where a value closer to 1 means a higher similarity. For vectors, UQI can capture the overall distribution and structural characteristics of the vectors. Therefore, in this thesis, UQI is used to enhance the completeness of similarity computation. ◦

Visual information fidelity (VIF)

Suppose that there is a given image, VIF [8] is to consider how much information we can get from the other image. In this thesis, we apply VIF to measure the similarity between vectors, considering the similarity of structure.

In VIF, the input data is different from the previous three methods. We form a 16×3 matrix by using the 16 original features from each player's first three seasons. For each 16 features, we take the 3-dimensional vector of the k -th statistic and compute the similarity between the rookie player and the historical player. Finally, we square the differences and sum from 16 features to generate our final similarity score. Therefore, for our VIF, the formula is defined as follows:

$$VIF(X_i, Y_j) = \sum_{k=1}^{16} \left\{ \frac{\log_2(1 + \frac{g_k^2 \sigma_{x_{i,k}}^2}{\sigma_{v_k}^2 + \sigma_{n_k}^2})}{\log_2(1 + \frac{\sigma_{x_{i,k}}^2}{\sigma_{n_k}^2})} - 1 \right\}^2$$

where:

- $g_k = \frac{x_{i,k}^T y_{j,k}}{x_{i,k}^T x_{i,k}}$.
- $\sigma_{v_k} = \frac{r_k^T r_k}{3}$, where $r_k = y_{j,k} - g_k x_{i,k}$.
- σ_{n_k} is the given coefficient.

A higher value means lower similarity between vectors, and if the two vectors are exactly the same, the value is 0. Compared to traditional distance difference measures such as WMSE

or PSNR, VIF can capture deeper structural similarities, giving a more effective similarity evaluation.

3.2.3 Combine

From 3.2.2, we have four similarity values, then we define the similarity of two players as follows:

$$\text{Similarity}(X_i, Y_j) = a \times \text{WMSE}(X'_i, Y'_j) + b \times \text{PSNR}(X'_i, Y'_j) + c \times \text{UQI}(X'_i, Y'_j) + d \times \text{VIF}(X_i, Y_j)$$

where a, b, c and d are coefficients.

Based on the above formulas, our idea is that the more similar two vectors are, the higher the final score should be. Therefore, we define a similarity score of 80 as the result of applying the four similarity measures by putting X'_i and $0.8X'_i$, and combining them linearly. Similarly, similarity scores of 60, 40, and 20 are defined using $0.6X'_i$, $0.4X'_i$ and $0.2X'_i$ respectively.

Therefore, we have the following five equations, and use them and the least squares method to solve the four coefficients in the linear combination.

$$\begin{cases} a \times \text{WMSE}(X'_i, 0.8X'_i) + b \times \text{PSNR}(X'_i, 0.8X'_i) + c \times \text{UQI}(X'_i, 0.8X'_i) + d \times \text{VIF}(X_i, 0.8X_i) = 80 \\ a \times \text{WMSE}(X'_i, 0.6X'_i) + b \times \text{PSNR}(X'_i, 0.6X'_i) + c \times \text{UQI}(X'_i, 0.6X'_i) + d \times \text{VIF}(X_i, 0.6X_i) = 60 \\ a \times \text{WMSE}(X'_i, 0.4X'_i) + b \times \text{PSNR}(X'_i, 0.4X'_i) + c \times \text{UQI}(X'_i, 0.4X'_i) + d \times \text{VIF}(X_i, 0.4X_i) = 40 \\ a \times \text{WMSE}(X'_i, 0.2X'_i) + b \times \text{PSNR}(X'_i, 0.2X'_i) + c \times \text{UQI}(X'_i, 0.2X'_i) + d \times \text{VIF}(X_i, 0.2X_i) = 20 \\ a \times 0 + b \times 33.7105 + c \times 1 + d \times 0 = 100 \end{cases}$$

After computing the coefficients, we have a combined similarity score ranging from negative infinity to 100. The score 100 means the two players are exactly the same, while lower values mean larger differences between them. A negative score suggests that the players are completely

different in type.

3.2.4 Prediction

From 3.2.3, we have the similarities between rookies and historical players, taking the top five for the prediction. Then, we define the prediction as follows:

$$Prediction = \frac{600 \times X + \sum_{i=0}^4 ((5-i) \times top5_i \times A_i)}{600 + \sum_{i=0}^4 ((5-i) \times top5_i)}$$

where:

- X is the rookie's average states of the first three seasons.
- $top5_i$ is the five largest numbers of $Similarity(X_i Y_j)$
- A_i is the average state of the corresponding players for 4-6 seasons.

Multiplying X by 600 is based on the idea that a player's development is similar to their own past performance. The number 6 comes from including the five most similar historical players, plus the rookie player himself as the sixth. Score 100 for himself, and thus receives the highest weight, while the remaining weights decrease according to similarity.

In this way, the prediction is not only base on the rookie's past performance but also consider the performance and similarity of comparable historical players, allowing for a more realistic prediction.

3.2.5 Adjustment

In this study [9], making final adjustments to the predicted values is considered important, it can reduce error and improve accuracy. Therefore, we notice that rookie players and historical

players have the following relationship, and we apply the following adjustment to the predicted values get above:

- $2PA \div 1.05$.
- $3PA \times 1.3$.
- $USG + 2$.
- $FGA = 2PA + 3PA$.
- $PTS = 2 \times 2PA \times 2P\% + 3 \times 3PA \times 3P\% + FTA \times FT\%$.
- Round $FG\%$, $2P\%$, $3P\%$ and $FT\%$ down to three decimal places.
- Round the others down to two decimal places.

There are 7 adjustments, from the first to the third, since in today's games, as the number of 3PA continues to rise, the number of 2PA decreases accordingly. In addition, offensive possessions are increasingly concentrated on star players, resulting in a noticeable increase in their USG. Fourth and fifth adjustments are common sense, we need to make our predictions reasonable. We need the last two adjustments since we want the predictions have the same decimal places as the reality data.

3.3 Method 2: Deep Learning model

Deep learning models can help us complete tasks more quickly and efficiently. Therefore, we want to use deep learning to predict the performance of NBA rookies. We chose the Multilayer Perceptron (MLP) as our model, as it has strong capabilities in data of vectors.

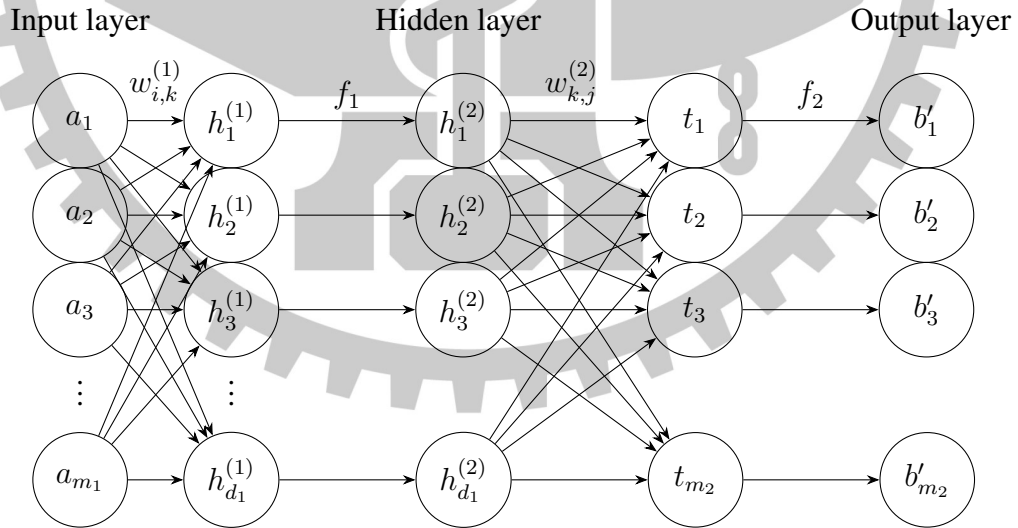
3.3.1 Normalization

Before applying deep learning, we also do normalization on the vectors to unify values with different features. For the set of all rookie and historical players, each player is represented as $A_i = \{a_{i,1}, a_{i,2}, \dots, a_{i,16}\}$, $i = 1, 2, \dots, 70$, the normalization is follows:

$$a'_{i,k} = \frac{a_{i,k} - \mu_k}{\sigma_k}, \quad i = 1, 2, \dots, 70, \quad k = 1, 2, \dots, 16$$

3.3.2 Multilayer Perceptron (MLP)

MLP is a type of feedforward neural network model composed of three main elements, including input layer, hidden layers and output layer. After computing the loss function, MLP uses back propagation (see derivation in the Appendix A.4 or [10]) to update gradients layer by layer to complete learning. Suppose we have sample set $\{A^{(i)}, B^{(i)}\}$ $i = 1, 2, \dots, n$, and $A^{(i)} \in \mathbb{R}^{m_1}, B^{(i)} \in \mathbb{R}^{m_2}$, the graph of MLP in follows :



From the input layer to the hidden layer, the data compute a weighted sum, followed by an activation function to get $h^{(2)}$. Then, another weighted computation is applied to get t , and applied by an activation function again to generate the final output. Therefore, we can express

the process as follows:

$$h_i^{(1)} = \sum_{k=1}^{m_1} w_{i,k}^{(1)} a_k \quad h_i^{(2)} = f_1(h_i^{(1)}) \quad t_j = \sum_{k=1}^{d_1} w_{j,k}^{(2)} h_k^{(2)} \quad b_j = f_2(t_j)$$

where:

- $w_{i,k}^{(1)}$ and $w_{j,k}^{(2)}$ are training weights.
- f_1 and f_2 are activation functions.

Activation functions help the MLP capture non linear features during training. The three most commonly used activation functions are ReLU, tanh, and sigmoid. Their definitions are as follows:

$$\text{ReLU: } f(x) = \max(0, x) \quad \text{Tanh: } f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \text{Sigmoid: } f(x) = \frac{1}{1 + e^{-x}}$$

Each function has different characteristics, and the choice of activation function affects the learning ability and convergence of the model.

In this thesis, we use the normalized data from the first three seasons of each historical player, connect them into a 48-dimensional input vector. This input is passed through a single hidden layer with 64 nodes. The learning target is the average performance of each historical player in their fourth to sixth seasons, represented as a 16-dimensional output vector. The activation function used is ReLU.

Since we only have 53 historical players, the training dataset is relatively small, which can lead to overfitting. Thus, we apply regularization in the MLP model [11] to improve its generalization ability. After training, we input the 17 target rookie players into the model to generate their predicted performance.

Chapter 4. Results

4.1 Comparison between two methods

In this thesis, our method is written in Python, and we use the Mean Absolute Percentage Error (MAPE) in L2-norm as our error function. For prediction vector Y' and real vector Y , the formula is defined by:

$$Error(Y', Y) = \frac{\|Y' - Y\|_2}{\|Y\|_2}$$

If the MAPE value is larger, it means the prediction is less accurate.

The following are MAPE of our two methods:

Table 1: MAPE Error for the two prediction models

Player	Similaity model	MLP
Shai Gileous-Alexander	0.2408	0.1473
Dejounte Murray	0.1089	0.1098
Trae Young	0.0716	0.0313
Luka Doncic	0.1018	0.0611
Fred VanVleet	0.1003	0.0719
De'Aaron Fox	0.0624	0.0821
Jalen Brunson	0.1071	0.0658
Jaylen Brown	0.074	0.1252
Donovan Mitchell	0.0472	0.0819
Jayson Tatum	0.0902	0.0792
Brandon Ingram	0.1229	0.1214
Pascal Siakam	0.0949	0.2134
Lauri Markkanen	0.1183	0.253
Bam Adebayo	0.1037	0.1057
Domantas Sabonis	0.1058	0.2314
Jarrett Allen	0.2403	0.1845
Jaren Jackson Jr.	0.3236	0.0844

From Table 1, the average MAPE are 0.1243 and 0.1205, and the standard deviations are 0.0625 and 0.0712, respectively. This shows that the two methods have strong predictive abilities, and they can achieve almost 88% accuracy. Additionally, most of the MAPE fall within one standard deviation from the mean. This shows the stability of our two methods.

For the smallest errors, the values are 0.0472 and 0.0313 with the corresponding players: Donovan Mitchell and Trae Young. We show the result of these two players:

Table 2: Result for Donovan Mitchell

Type	PTS	REB	AST	STL	BLK	TOV	PF	FG%	FGA	2P%	2PA	3P%	3PA	FT%	FTA	USG
Reality	27.9	4.6	5.7	1.3	0.2	3.1	2.5	0.444	22	0.507	12.1	0.368	9.9	0.849	5.6	32.8
Similarity model	26.6	4.4	4.7	1.0	0.2	3.0	2.5	0.459	20.8	0.507	11.9	0.368	8.9	0.851	5.4	32.0

Table 3: Result for Trae Young

Type	PTS	REB	AST	STL	BLK	TOV	PF	FG%	FGA	2P%	2PA	3P%	3PA	FT%	FTA	USG
Reality	27.5	3.3	10.4	1.1	0.1	4.2	1.7	0.442	19.9	0.491	12.1	0.365	7.8	0.884	8.1	32.7
MLP model	28.4	3.3	9.7	0.9	0	4.2	1.9	0.461	20.5	0.512	12.4	0.347	8.0	0.870	8.3	31.9

Now, we consider some large errors in the two methods and find the reasons. For Shai Gileous-Alexander, the two methods have relatively large errors because if we see his data (see Table 4), he has a big outbreak in his four to six seasons, his average PTS grows from 19 points to 30 points. This is a rare thing in NBA history and we under rate the prediction of Shai Gileous-Alexander.

Table 4: Statistics for Shai Gileous-Alexander

Type	PTS	REB	AST	STL	BLK	TOV	PF	FG%	FGA	2P%	2PA	3P%	3PA	FT%	FTA	USG
First three	19	5	4.4	1.3	0.7	2.4	2.3	0.482	14.3	0.571	10.8	0.373	3.5	0.805	4.9	22.5
4-6 seasons	30	5.3	6.1	1.7	0.9	2.6	2.7	0.504	20.4	0.545	16.6	0.33	3.8	0.872	9.3	32.2
Similarity model	21.9	4.4	6.2	1.2	0.4	2.7	2.1	0.463	16.9	0.496	10.7	0.365	6.2	0.83	5.3	27.0
MLP	24.4	6.5	6.0	1.2	0.7	3.2	2.2	0.477	17.9	0.514	12.9	0.338	5.0	0.818	7.3	30.2

Additionally, there are relatively large errors from Pascal Siakam to Jaren Jackson Jr. Since they are power forwards or centers, these two positions have a big change in playing style than other positions. Big men are no longer limited to staying near the basket; they develop more shooting abilities to help create offensive spacing. In many cases, they also become the main points in offensive tactics. However, among the historical big men we selected, few have this change. As a result, when predicting the performance of rookie players, this leads to larger prediction errors.

4.2 Comparison with other methods

In [2], there are Linear Regression and Support Vector Regression (SVR). We put our data into the algorithm and compare the result with the similarity measures model and the MLP. We show the MAPE result of the features: PTS, REB and AST:

Table 5: MAPE for four methods in PTS, REB and AST

Methods	PTS	REB	AST
Linear Regression	0.0802	0.1081	0.1915
SVR	0.0892	0.1077	0.1467
Similarity model	0.0859	0.1593	0.1961
MLP model	0.1088	0.1396	0.1351

From the Table 5, it shows that our two methods have a good performance on predicting PTS, REB and AST. Especially, in PTS and AST respectively, the MAPE errors can reach 0.0859 and 0.1351.

Chapter 5. Conclusions

This thesis focuses on 17 high potential rookie players who entered the NBA between 2016 and 2018 and have at least one All-Star appearance. For historical players, we selected 53 All-Star players from the 2015–2024 period. Each player’s 16 traditional statistics from their first three seasons are normalized and then put into two different methods: the ensemble similarity measures model and the Multilayer Perceptron (MLP) model.

The ensemble similarity measures model combines four similarity measures, including WMSE, PSNR, UQI and VIF, then uses the least squares method to compute the weights. The prediction is computed by a weighted average of the five most similar historical players. The MLP model takes a 48-dimensional vector as input data and uses a hidden layer with 64 nodes. Regularization is applied to prevent overfitting. Learning the mapping between a player’s first three years and their performance in the following three seasons.

Results show that the MAPE errors of rookie players’ prediction are 0.1243 and 0.1205, and the standard deviations are 0.0625 and 0.0712, respectively. It is equivalent to an overall accuracy of approximately 88%, and our two methods are stable. Additionally, compared to Linear Regression and SVR, our two methods have good performance and in PTS and AST especially.

For the individual cases, Trae Young and Donovan Mitchell have the lowest MAPE errors, 0.0313 and 0.0472. However, players like Shai Gilgeous-Alexander and several big men such as Pascal Siakam show larger MAPE errors due to performance outbreak and the different playing style between rookie and historical players.

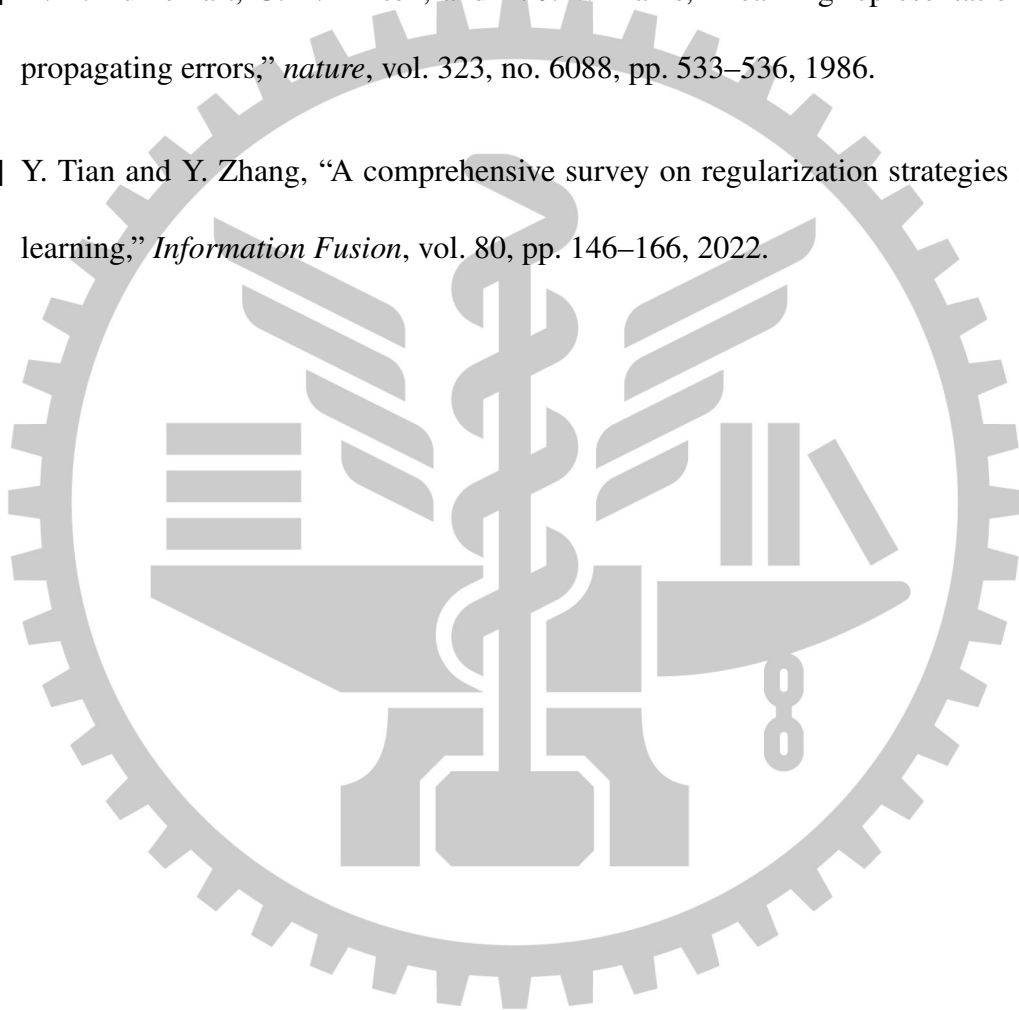
For the future work, there are two directions. First, we want to consider more aspect for big men such that we can make a more accurate prediction for these types of players. Second, we want to relax the condition in players' selection, then we can have more players in our data. It can increase the generalization ability of our models.



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Appendix

Table A.1: 16 Features and Corresponding Weights

Feature	Explanation	Weight
PTS	Points per 36 minutes	15
REB	Rebounds per 36 minutes	15
AST	Assists per 36 minutes	6
STL	Steals per 36 minutes	6
BLK	Blocks per 36 minutes	5
TOV	Turnovers per 36 minutes	5
PF	Personal Fouls per 36 minutes	6
FG%	Field Goal Percentage	5
FGA	Field Goal Attempts	2
2P%	2-Point Percentage	5
2PA	2-Point Attempts	2
3P%	3-Point Percentage	5
3PA	3-Point Attempts	2
FT%	Free Throw Percentage	3
FTA	Free Throw Attempts	3
USG%	Usage Percentage	7

Table A.2: 17 Rookie Players

Player	Position
Shai Gileous-Alexander	PG
Dejounte Murray	PG
Trae Young	PG
Luka Doncic	PG
Fred VanVleet	PG
De' Aaron Fox	PG
Jalen Brunson	PG
Jaylen Brown	SG
Donovan Mitchell	SG
Jayson Tatum	SF
Brandon Ingram	SF
Pascal Siakam	PF
Lauri Markkanen	PF
Bam Adebayo	C
Domantas Sabonis	C
Jarrett Allen	C
Jaren Jackson Jr.	C

Table A.3: 53 Historical Players

Player	Position	Player	Position
Kyle Lowry	PG	Kevin Durant	SF
John Wall	PG	Kawhi Leonard	SF
Isaiah Thomas	PG	Gordon Hayward	SF
Stephen Curry	PG	Khris Middleton	SF
Russell Westbrook	PG	Andrew Wiggins	SF
Chris Paul	PG	Paul Millsap	PF
Kyrie Irving	PG	Pau Gasol	PF
Kemba Walker	PG	LaMarcus Aldridge	PF
Damian Lillard	PG	Anthony Davis	PF
Ben Simmons	PG	Draymond Green	PF
D'Angelo Russell	PG	Giannis Antetokounmpo	PF
Mike Conley	PG	Kevin Love	PF
Jrue Holiday	PG	Kristaps Porzingis	PF
Dwyane Wade	SG	Blake Griffin	PF
Jimmy Butler	SG	Dirk Nowitzki	PF
DeMar DeRozan	SG	Julius Randle	PF
Kobe Bryant	SG	Chris Bosh	C
James Harden	SG	Andre Drummond	C
Klay Thompson	SG	Al Horford	C
Bradley Beal	SG	DeMarcus Cousins	C
Victor Oladipo	SG	Marc Gasol	C
Devin Booker	SG	DeAndre Jordan	C
Zach Lavine	SG	Rudy Gobert	C
Camelo Anthony	SF	Karl-Anthony Towns	C
Paul George	SF	Nikola Vucevic	C
LeBron James	SF	Nikola Jokic	C
		Joel Embiid	C

A.4 Derivation of backward propagation

Suppose we have real value $B^{(i)} = \{b_1^{(i)}, b_2^{(i)}, \dots, b_{m_2}^{(i)}\}$ and

prediction value $B'^{(i)} = \{b_1'^{(i)}, b_2'^{(i)}, \dots, b_{m_2}'^{(i)}\}$, we compute loss function $E^{(i)}$ by Mean Square

Error, then we sum the all of them. Therefore, we can get:

$$E^{(i)} = \frac{1}{m_2} \sum_{k=1}^{m_2} (b_k^{(i)} - b_k'^{(i)})^2 \quad E = \sum_{i=1}^n E^{(i)}$$

Our goal is to find the training weights such that we can get a minimum of E :

$$\arg \min_{w_{i,k}^{(1)}, w_{k,j}^{(2)}} \{E\}$$

By ODE, we know that the solution needs to satisfy the following conditions:

$$\frac{\partial E}{\partial w_{i,k}^{(1)}} = 0 \quad \frac{\partial E}{\partial w_{k,j}^{(2)}} = 0$$

In Deep Learning, the gradient descent is an important tool, and we use it to help us solve the equation, for η is our learning rate, the gradient descent will be:

$$w_{i,k}'^{(1)} = w_{i,k}^{(1)} - \eta \Delta w_{i,k}^{(1)} \quad w_{k,j}'^{(2)} = w_{k,j}^{(2)} - \eta \Delta w_{k,j}^{(2)}$$

where:

- $\Delta w_{i,k}^{(1)} = \frac{\partial E}{\partial w_{i,k}^{(1)}} = \sum_1^n \frac{\partial E^{(i)}}{\partial w_{i,k}^{(1)}}$
- $\Delta w_{k,j}^{(2)} = \frac{\partial E}{\partial w_{k,j}^{(2)}} = \sum_1^n \frac{\partial E^{(i)}}{\partial w_{k,j}^{(2)}}$

Now, we split the partial differential by the chain rule, then we can get that:

$$\Delta w_{i,k}^{(1)} = \sum_1^n \frac{\partial E^{(i)}}{\partial w_{i,k}^{(1)}} = \sum_1^n \sum_{j=1}^{m_2} (b_j^{(i)} - b_j'^{(i)}) f_2'(t_j^{(i)}) h_k^{(2)}$$

$$\Delta w_{k,j}^{(2)} = \sum_1^n \frac{\partial E^{(i)}}{\partial w_{k,j}^{(2)}} = \sum_1^n \sum_{j=1}^{m_2} (b_j^{(i)} - b_j'^{(i)}) f_2'(t_j^{(i)}) w_{i,k}^{(1)} f_1'(h_k^{(1)}) a_i$$

As a result, if we have a loss function, then we can compute the training weight we want. □

